



31-MAVZU · III BOB. PYTHON VA MA'LUMOTLAR BAZASI

# Yakuniy nazorat ishi — II–III boblar (JavaScript, Python, SQL)

JavaScript, Python va ma'lumotlar bazasi bo'yicha nazorat: 25 ta yopiq test (3 balldan) va 5 ta ochiq savol/amaliy topshiriq (5 balldan) — jami 100 ball. Takrorlash rejasi, javoblar va baholash mezonlari.

# Nazorat ishi qanday o'tadi?

| Qism        | Mazmuni                                           | Ball       |
|-------------|---------------------------------------------------|------------|
| 1-qism      | 25 ta test × 3 ball (JS — 8, Python — 9, SQL — 8) | 75         |
| 2-qism      | 5 ta ochiq savol / amaliy topshiriq × 5 ball      | 25         |
| <b>Jami</b> | Vaqt: 80 daqiqa                                   | <b>100</b> |

- Bugun: takrorlash → namunaviy testlar → amaliy topshiriqlar → baholash

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# Takrorlash

II va III boblarning eng muhim tushunchalari

# JavaScript: 15–20-mavzular

| Mavzu             | Bilishingiz kerak                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JS asoslari       | <code>let/const</code> , turlar, <code>typeof</code> , <code>===</code> , <code>if/else</code> , <code>switch</code> , ternary                                              |
| Sikl, funksiya    | <code>for</code> , <code>while</code> , <code>break</code> , <code>function</code> , <code>return</code> , arrow                                                            |
| Massiv,<br>obyekt | <code>push/pop</code> , <code>map/filter/reduce</code> , <code>obj.kalit</code>                                                                                             |
| DOM,<br>hodisalar | <code>querySelector</code> , <code>textContent</code> , <code>classList</code> , <code>addEventListener</code> , <code>preventDefault</code> ,<br><code>localStorage</code> |

# Python va SQL: 21–30-mavzular

| Mavzu              | Bilishingiz kerak                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Python asoslari    | <code>print/input</code> , <code>//</code> <code>%</code> , f-satr, <code>if/elif</code> , <code>for + range</code> , <code>try/except</code> , fayllar             |
| Tuzilmalar, OOP    | <code>list/tuple/set/dict</code> , slicing, <code>def</code> , modullar, <code>json</code> ; <code>class</code> , <code>__init__</code> , <code>self</code> , meros |
| Ma'lumotlar bazasi | PK/FK, bog'lanishlar, turlar, <code>CREATE</code> , CRUD                                                                                                            |
| SQL chuqur         | <code>JOIN</code> , indeks, <code>GROUP BY</code> , <code>HAVING</code>                                                                                             |

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# Namunaviy testlar

Avval o'ylang — keyin javobni ko'ring

# Savol

---

"3"  $\equiv$  3 natijasi?

**A** true

**B** "33"

**C** 6

**D** false

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

`[5, 10, 15].map(x ⇒ x / 5)` natijasi?

**A** 30

**B** `[5, 10, 15]`

**C** `[1, 2, 3]`

**D** 6

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

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Massivni localStorage'ga saqlashdan oldin nima qilinadi?

**A** `Number()`

**B** `toString()`

**C** Hech narsa

**D** `JSON.stringify()`

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

`list(range(1, 5))` natijasi?

**A** `[1, 2, 3, 4, 5]`

**B** `[0, 1, 2, 3, 4]`

**C** `[1, 5]`

**D** `[1, 2, 3, 4]`

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

`d = {"a": 1}; print(d.get("b", 0))` natijasi?

**A** KeyError

**B** None

**C** 0

**D** 1

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

Pul summasi uchun eng mos tur?

**A** NUMERIC(12, 2)

**B** REAL

**C** VARCHAR

**D** INTEGER

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

**Chap jadvalning barcha qatorlarini qaytaruvchi JOIN?**

**A** INNER JOIN

**B** RIGHT JOIN

**C** LEFT JOIN

**D** CROSS JOIN

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

Guruhlangan natijani shart bo'yicha filtrlash?

**A** WHERE

**B** HAVING

**C** FILTER

**D** IF

Javobni tanlang — to'g'ri variant yashil bo'ladi

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# Amaliy qism

5 ta topshiriq × 5 ball

# 1-topshiriq: JavaScript DOM

- Tugma bosilganda: maydon bo'sh — "Ismingizni kiriting!" (qizil), aks holda — "Salom, [ism]!" (yashil)

JavaScript

```
document.querySelector("#btn").addEventListener("click", () => {  
  const ism = document.querySelector("#ism").value.trim();  
  const javob = document.querySelector("#javob");  
  if (ism === "") {  
    javob.textContent = "Ismingizni kiriting!";  
    javob.style.color = "red";  
  } else {  
    javob.textContent = `Salom, ${ism}!`;  
    javob.style.color = "green";  
  }  
});
```

## 2-3-topshiriqlar: Python

### 2: statistika

```
def statistika(ballar):  
    if not ballar:  
        return None  
    return {  
        "ortacha": round(sum(ballar) / len(ballar), 2),  
        "eng_yuqori": max(ballar),  
        "eng_past": min(ballar),  
        "alo": len([b for b in ballar if b > 86]),  
    }
```

### 3: OOP

```
class Tortburchak:  
    def __init__(self, eni, boyi):  
        self.eni = eni  
        self.boyi = boyi  
    def yuza(self):  
        return self.eni * self.boyi  
    def perimetr(self):  
        return 2 * (self.eni + self.boyi)  
    def __str__(self):  
        return f"Tortburchak {self.eni}x{self.boyi}"  
  
class Kvadrat(Tortburchak):  
    def __init__(self, tomon):  
        super().__init__(tomon, tomon)
```

## 4-topshiriq: SQL jadval va CRUD

SQL

```
CREATE TABLE kitoblar (  
    id        SERIAL PRIMARY KEY,  
    nomi      VARCHAR(200) NOT NULL,  
    muallif   VARCHAR(100),  
    yil       INTEGER CHECK (yil > 1450),  
    narx      NUMERIC(10, 2),  
    mavjud    BOOLEAN DEFAULT TRUE  
);  
INSERT INTO kitoblar (nomi, muallif, yil, narx) VALUES  
    ('O'tkan kunlar', 'Abdulla Qodiriy', 1926, 65000),  
    ('Ufq', 'Said Ahmad', 1976, 48000);  
SELECT nomi, yil, narx FROM kitoblar WHERE yil > 2000 ORDER BY narx DESC;  
UPDATE kitoblar SET narx = narx * 1.1 WHERE id = 1;
```

## 5-topshiriq: JOIN va GROUP BY

- Har sinf: nomi, o'quvchilar soni, o'rtacha baho; faqat o'rtachasi 4 dan yuqorilar

SQL

```
SELECT s.nomi,  
       COUNT(DISTINCT o.id) AS oquvchilar,  
       ROUND(AVG(b.baho), 2) AS ortacha  
FROM sinflar s  
JOIN oquvchilar o ON o.sinf_id = s.id  
JOIN baholar b   ON b.oquvchi_id = o.id  
GROUP BY s.nomi  
HAVING AVG(b.baho) > 4  
ORDER BY ortacha DESC;
```

- Nega `COUNT(DISTINCT o.id)` ? JOIN'dan keyin o'quvchi har bahosi uchun takrorlanadi!

# Baholash shkalasi

| Ball   | Natija     | Tavsiya                                            |
|--------|------------|----------------------------------------------------|
| 86–100 | A'lo       | IV bob — Django'ga tayyorsiz!                      |
| 71–85  | Yaxshi     | Xato qilgan mavzular amaliyotini takrorlang        |
| 60–70  | Qoniqarli  | OOP (25) va JOIN/GROUP BY (29–30) ni qayta ishlang |
| 0–59   | Qoniqarsiz | Mentor bilan qo'shimcha mashg'ulot                 |

## III bob yakunlandi!

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- Ikki dasturlash tili: **JavaScript** va **Python**
- **OOP** tamoyillari
- Ma'lumotlar bazasini loyihalash va **SQL** bilan boshqarish

**Keyingi bob: Python + PostgreSQL + Django = haqiqiy veb-ilova!**



# Savollar bormi?

Keyingi mavzu: **32. Python dasturlash tilini PostgreSQL ma'lumotlar bazasiga bog'lash**

Batafsil matn, misollar va testlar: **31-nazorat-ishi-2.html** darsida